Project 5 : LDR Module

Introduction:

Pluto X2, LDR module is one of the add-ons. Which detects the change in environment.

Challenge:

Build a program for drone that lands when it enters environments with low illuminance.

Add-On used:

An LDR (Light Dependent Resistor) is a sensor that detects light intensity**.**
Its resistance changes depending on intensity of light falling on it.
More light → lesser resistance
Less light → more resistance

Connection:

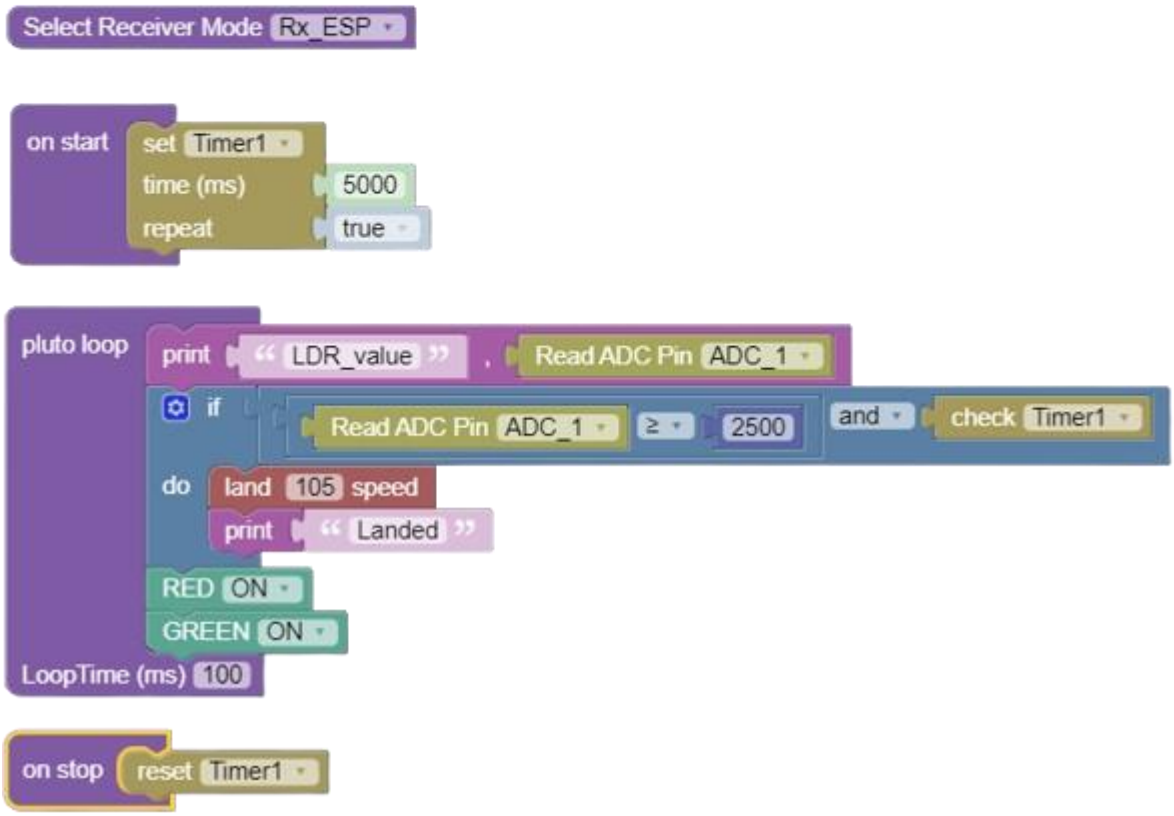
Connect the LDR sensor on the ADC port of Primus V5/X2 board

Primus V5/X2 pin	Peripheral
VCC on ADC port	VCC on LDR Sensor
GND on ADC port	GND on LDR Sensor
Signal on ADC port	A0 on LDR Sensor

Approaching the Problem:

We utilise a timer, to create a period of 5 seconds. The decision to land will depend on the LDR readings and timer1. It is necessary to use a timer here of atleast 3 seconds, because the time taken to land depends upon the height it is hovering. In most cases 3-4 seconds is normal duration to land with default speed of 105.

Solution:



Explaination of Solution:

On Start Loop:

We set a repeatable timer of 5 seconds

Pluto Loop:

Indication of Pluto Loop running is glowing of yellow light on its indicator LED. At ADC port, LDR sensor returns value from 0 to 4096. The more the illumination, lesser will be the reading. So, when drone enters a dim room, the reading goes above 2500 magnitude. The time taken by drone to land at default landing speed(105) is 3-5 seconds. Harnessing these facts, we have designed our code to land whenever it encounters dim light on LDR sensor.

On stop Loop: Timer1 resets.

fig: LDR addon with Pluto

Try further!

- Make a Light flash follower drone.
- Add a another LDR and try making a Wall line follower drone.